

Lirui Wang

5000 25th Ave NE, Seattle, WA 98105

✉ liruiw@uw.edu

🌐 liruiw.github.io

☎ +1 (206) 483-8515

EDUCATION

University of Washington, Seattle, WA

Bachelor of Science, Electrical Engineering, (Honors), College of Engineering

Bachelor of Science, Computer Science, (Honors), College of Arts and Sciences

Bachelor of Science, Mathematics, (Minor), College of Arts and Sciences

Sep. 2016 – Jun. 2020

GPA: 3.97 / 4.00, Advisor: Prof. Dieter Fox

RESEARCH PROJECTS

Manipulation Trajectory Optimization with Online Grasp Synthesis and Selection, Seattle, WA

UW Robotics and States Estimation Lab

March. 2019 – Present.

- Prototype RL, IL, and MPC with PyBullet environment to map observed points directly to control in task space.
- Implement CHOMP Planner and introduce an integrate planner that optimizes manipulation trajectory with online grasp synthesis and selection.

Survey and Empirical Evaluation of Attacks and Defense for Robust Classification, Seattle, WA

Robust Deep Learning Course Project

Sep. 2019 – Dec. 2019

- Investigate example reweighting and adversarial training techniques to robustify a fossil classification task and survey recent papers about attacks and defense of adversarial examples for neural networks.

Machine Learning in Robot Trajectory Optimization, Seattle, WA

Statistical Learning Course Project

Sep. 2019 – Dec. 2019

- Summarized the formulation of trajectory optimization as functional gradient descent using Reproducing Kernel Hilbert Space (RKHS) and as probabilistic inference using Gaussian Process (GP).

6D Object Pose Tracking with the Deep Iterative Matching Network, Seattle, WA

UW Robotics and States Estimation Lab

June. 2018 – March. 2019

- Investigated DeepIM on YCB-Video datasets and achieved robust results trained on synthetic data only
- Extended DeepIM in real-time tracking robot arm and objects for occluded scenes in YCB-Video dataset and built AR demos

Unsupervised Flow and Depth Methods Study, Seattle, WA

UW Robotics and States Estimation Lab

Jan. 2018 – Jun. 2018

- Investigated a few unsupervised learning method such as SfMlearner and MonoDepth for 3D geometry tasks such as depths, flow, and egomotion on a robotic setting

Baxter Hand-Eye Coordination, Seattle, WA

UW Robotics and States Estimation Lab

Aug. 2017 – Jan. 2018

- Worked on camera models and hand-eye calibration, and integrate PoseCNN to allow Baxter to manipulate daily objects.
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EXPERIENCE

Teaching Assistant, Seattle, WA

UW CSE 455 (Computer Vision)

Sep. 2019 – Present.

Research Assistant, Seattle, WA

UW Adaptive Computing Machines and Emulator Lab

March. 2018 – Jun. 2018

- Designed a sound processing pipeline from scratch for modelling the acoustics of a train whistle

Computer Vision Lead & Drone Lead, Seattle, WA

Advanced Robotics at the University of Washington

Sep. 2016 – Feb. 2018

- Managed a team of 10 students to design computer vision tools for our robots in a competition RoboMasters hosted by DJI
- Perform system integration of controls, navigation and serial communication RoboMasters hosted by DJI

Robotics Engineer Intern, Shenzhen, China

DJI Technology

Jun. 2017 – Aug. 2017

- Worked on automatic robot battle of vehicles with TX series onboard and DJI Manifold with Guidance
 - Applied SLAM to mapping and localize, and built a top-level state machine to strategize and navigate the robot
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SKILLS & OTHERS

Programming Language: Java, C++/C, Python, Matlab

Frameworks: ROS, Ubuntu, ARM, Arduino, Nvidia TX Series, STM32

Tools: Git, CMake, OpenCV, DJI Guidance, Bash, Latex, Tensorflow, Pytorch, MXNet, OpenGL, CUDA, Qt

Courses: Computer Vision, Deep Learning, Artificial Intelligence, Graphics, Robotics, Algorithms, Numerical Analysis, Linear Analysis, Convex Analysis and Optimization, Embedded Systems, Control Theory, Digital Signal Processing, Statistical Learning, High-Dimension Statistics, Robust Machine Learning

Scholarships: Tau Beta Pi Scholarship 2017-2019, Lawrence & Lucille Frey Endowed Electrical Engineering Scholarship 2017-2018, Patricia G. Lynch and Theodora & Eugene Russell Memorial Computer Science Scholarship 2018-2019